



北京邮电大学



EBU5408 Digital Audio Fundamentals

Lab 2

Introduction

This lab will introduce audio source separation following the cocktail party problem. Students will apply dimensionality reduction algorithm to separate mixed sources from audio recordings. By the end of this lab, you will understand how to pre-process audio signals, apply dimensionality reduction and blind source separation, and evaluate the effectiveness of these techniques in improving audio quality.



The tasks of this lab use PYTHON.

Learning Objectives

- Load and preprocess multi-channel audio recordings
- Apply techniques/algorithm to separate sources
- Visually, analyse, evaluate and optimise performance



All Lab deliverables must be submitted to QM+ **BEFORE** the lab session.

Lab attendance is mandatory. During the lab: interact with your TA and answer their questions about your lab deliverables (15%).

Lab 2 Submission to QM+

- Save in a “results folder” all the lab deliverables including your Python programming outcomes (e.g. the Python .py files and generated plot figures).
- Name your folder: ‘Lab2_EBU5408_xxxxxxx’ where xxxxxxxx is your QM student number.
- Upload your folder as a zip archive to QM+ in the EBU5408 course area **by 22nd May (firm deadline).**

No submission of the Lab deliverables will be accepted after this deadline.

No submission of the Lab deliverables will result in a mark of 0 for the Lab.

IMPORTANT: Plagiarism (copying from other students or copying the work of others without proper referencing) is cheating and **will not be tolerated**. IF TWO “FOLDERS” ARE FOUND TO CONTAIN IDENTICAL MATERIAL, BOTH WILL BE GIVEN A MARK OF ZERO.



The coursework report is to be submitted after Lab 2.

Getting Started

Download all the resources needed for the lab (i.e. audio files) in “lab2” from QMPlus.

Lab 2 – Audio Source Separation

Part A: Data Acquisition and Preprocessing

Task Overview:

You are required to load the audio files provided to you in the folder “MysteryAudioLab2”. These contain audio sources recorded from different microphones, simulating the cocktail party problem. You need to perform initial data inspection for this task.



1. Load the audio files provided in the folder “MysteryAudioLab2”.
2. Inspect the signals (sampling rate, duration, number of channels).
3. Plot the waveforms of the mixed signals.
4. Save your plots.



- D1: Waveform and spectrogram plots of the mixed signals
- D2: Python (.py) file used for loading and inspection



- Answer questions Lab2 **Q1** and **Q2** in the CW report.

Part B: Algorithm Implementation



1. Select and implement an appropriate algorithm to separate the preprocessed audio sources.
2. Apply the algorithm to the preprocessed signals.
3. Save the separated/estimated signals.
4. Plot the waveforms of the separated signals.
5. Generate spectrograms of the separated signals.
6. Save all plots.



- D5: Python (.py) implementing the separation algorithm
- D6: Waveform plots of separated signals
- D7: Spectrograms of separated signals/sources



- Answer questions Q3, Q4 and Q5 in the CW report.

Part C: Parameter Tuning and Evaluation



Task Overview:

You need to iteratively adjust key parameters such as components and any other parameters (if any) to optimise separation quality.

1. Adjust key parameters of your selected algorithm.
2. Re-run the separation with different parameter settings.
3. Compare the results obtained under different configurations.
4. Save plots and any evaluation results.



- D8: Plots demonstrating the impact of parameter changes
- D9: Any metrics or additional evaluation outputs



- Answer questions Q6 in the CW report.

Part D: Create your own Audio



Task: Record your own audio clip with different sources, such as other people talking. Apply source separation techniques to separate your voice from the other sources. The audio must be complex, with multiple sources (e.g. multiple people talking in the background with music or any other noise). Simple recordings with no or minimal background noise will not get mark for this task. This is to experiment creatively while demonstrating your understanding of audio processing techniques.

1. Record a real-world audio clip of your voice containing multiple simultaneous background sound sources.
2. Apply your source separation method to the recorded mixture.
3. Save the original recording and the separated signals.
4. Generate waveform and spectrogram plots of your results.



- D10: Original recorded mixture
- D11: Separated signals
- D12: Supporting waveform and spectrogram plots



- Answer questions Q7 and Q8 in the CW report.